

Autonomous Software Agents: Deliveroo.js

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Our Solution

- **BDI core**: sense $\rightarrow$ deliberate $\rightarrow$ plan $\rightarrow$ act; collects, explores, navigates
- **PDDL planner**: pushes crates when A* is blocked
- **LLM layer**: interprets mission commands; injects goals, rules, penalties
- **Coordination**: 2PC protocols, peer-injection, team coordinator

Beliefs: Tracker

- Stores latest observation $\langle v, t_{seen} \rangle$ per entity
- Updated on each sensing report
- Invalidated when entity is visible-but-absent

Confidence decay

$$c(t) = \exp\left(-\frac{t - t_{seen}}{\tau}\right)$$

Beliefs: Memory

- Bounded temporal history of observations per entity
- Entries older than a configurable TTL are evicted automatically
- Tracks enemy position history for heat-field computation

Beliefs: Agents

- **Self**: own position and current state
- **Friends**: latest state per teammate via Tracker
- **Enemies**: latest state via Tracker; position history via Memory
- **EnemyPredictor**: estimates each enemy's next position from history
- **Spatial heat field**: aggregates past enemy observations across the map

Beliefs: Map & Crates

- Static tile layout, transient blocked tiles, traversal penalties
- Crates tracked as dynamic obstacles; latest position via Tracker;

Beliefs: Map - Safe Zones

Directional Tiles create one-way edges, may trap agents in dead zones

Safe SCC (Tarjan's algorithm)

- Map is represented as a directed graph with edges for valid movements;
- Tarjan's algorithm identifies SCCs in this graph;
- An SCC is *safe* iff it contains ≥ 1 spawn tile *and* ≥ 1 delivery tile;
- All tiles outside safe SCCs are reclassified as walls.

Reminder: an SCC (strongly connected component) is a maximal set of nodes where every node is reachable from every other along directed edges.

Beliefs: Map - Safe Zones

[Safe SCC demo]

Beliefs: Parcels

- Reward decay is applied even when out of sensor range
- Parcel p removed from beliefs when $r_p(t) \leq 0$

Estimated reward

$$r_p(t) = r_{p,0} - \left\lfloor \frac{t - t_0}{\Delta_t} \right\rfloor$$

Desire: Distance Metric

- Manhattan distance poorly correlates with actual path;
- BFS from target tile computes true shortest path distance accounting for walls and crates;
- This gives a more accurate heuristic for desire scoring.

Desire: ReachParcel

- Navigate to and collect a visible parcel
- Priority 1: competes directly with DeliverParcel

Score

$$S = c_p \cdot \frac{r_p + \sum_{c \in C} r_c}{d} \cdot \gamma, \quad \gamma = \min\left(1, \frac{d_{enemy}}{d}\right)$$

- r_p : parcel reward;
- $\sum_{c \in C} r_c$: value of carried stack;
- c_p : confidence that parcel is still at its last known position;
- γ : race factor that penalizes parcels where enemies have advantage;

Desire: DeliverParcel

- Navigate to delivery tile while carrying parcels
- Priority 1: competes directly with ReachParcel

Score

$$S = \frac{\sum_{c \in C} r_c}{d}$$

Desire: Explore

- Seek unvisited spawn tile when no parcels are reachable
- Priority 0: fallback;
- w : cluster density; τ : freshness; h : enemy heat

Score

$$S = \frac{w \cdot \tau}{d \cdot (1 + h)}$$

Desire: Explore - Variables

Freshness τ

$$\tau = \min\left(1, \frac{\Delta t}{T_{\text{spawn}}}\right)$$

Δt : time since spawn tile last observed; T_{spawn} : spawn interval

Cluster weight w

$$w = \sum_{y \in OST} \frac{1}{\delta(x, y) + 1}$$

OST : spawn tiles within observation range; $\delta(x, y)$: BFS distance from x to y

Enemy heat h

$$h = \sum_{e \in E} \exp\left(-\frac{d_e^2}{2\sigma^2}\right) \exp\left(-\frac{\Delta t_e}{\tau_h}\right)$$

Spatial $\times$ temporal decay over tracked enemy observations

Collision: Detection

Checked before every move step. Target tile is *occupied* if:

- An enemy is currently observed on it, *or*
- An enemy's *predicted* next position matches it with sufficient confidence

Enemy position prediction

Next position estimated from last known state and movement history; only predictions above a confidence threshold trigger a block.

Collision: Escalation

- ① **Detour:** replan with tile as wall;
accepted only if extra steps $\leq \Delta_{th}$
- ② **Wait:** no viable detour:
pause for random duration $\sim \mathcal{U}$
- ③ **Block & replan:** after wait expires
or after L repeated invalidations:
mark tile blocked for random TTL, full
replan

[collision resolution]

Collision with Friends

- The escalation above applies only to enemies;
- Asymmetric by agent id: the *lower-id* agent yields;

When A* Fails

- A* fails: no path that avoids crates
- Second A* ignoring crates finds a “ghost” path
- Crates intersecting that path are part of the “PDDL” problem instance

PDDL

- Cluster detected → PDDL problem built from current beliefs
- Solver finds move + push sequence to reach goal
- Result translated back to agent's action plan

[PDDL demo]

Performance Bottlenecks

- **Real-Time Constraints:** The initial PDDL solver integration was too slow to meet the game's strict timing demands.
- **Identified Issues & Fixes:**
 - **API Overhead:** Remote solver calls introduced heavy latency. **Fix:** Migrated to a local solver instance.
 - **Redundant Computation:** Identical crate configurations were being solved repeatedly, but implementing a standard cache was non-trivial.
- **The Pivot:** Rather than brute-forcing a complex caching mechanism, we restated the problem to naturally maximize cache hits.

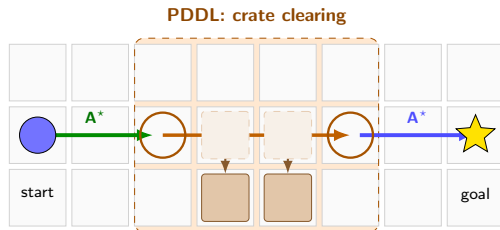
Optimization: Problem Restatement

Three-Phase Journey Segmentation

- ① **Approach:** A* pathfinding from the current position to the cluster *entry*.
- ② **Manipulation:** PDDL execution solely through the crate cluster (entry → exit).
- ③ **Completion:** A* pathfinding from the cluster *exit* to the final goal.

Optimization: Problem Restatement

- **The Result:** The PDDL problem is now strictly anchored to the cluster entry tile rather than absolute map coordinates.
- **Simplified Cache Key:** $\langle \text{entry, exit, \{crate positions\}} \rangle$.



LLM Layer Overview

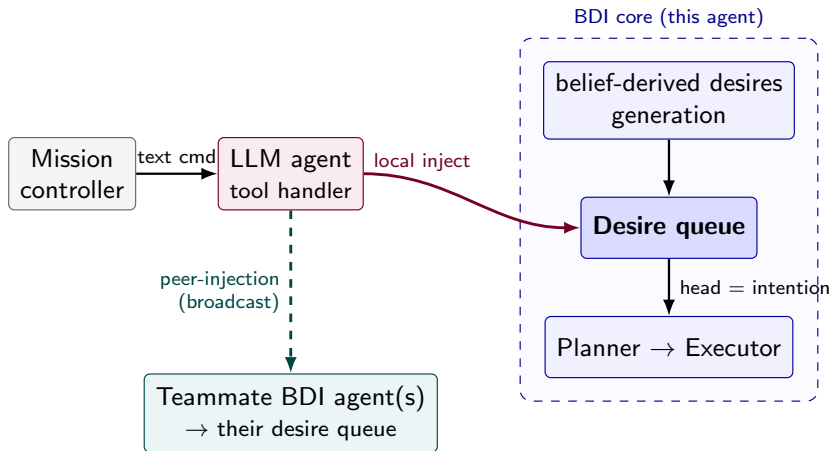
- Thin wrapper around the BDI core
- Listens and interprets mission-controller chat messages
- Modifies the BDI behaviour accordingly
- Four command categories: **goal injection**, **rules**, **coordination** and **cooperation**

Goal Injection

Trigger Examples

- *“Move to coordinate (4,7) and you get +10pts”*
 - *“Drop a package in the leftmost tile to get 5pt”*
-
- LLM can inject new goals into the BDI agent (e.g. go-to, deliver-at)
 - Compete with existing BDI desires; may be accepted or rejected
 - Persist and re-evaluated until executed or TTL expires
 - **Peer-injection:** the LLM applies the command locally, then repackages it and broadcasts it

Goal Injection: Local + Peer Broadcast



Rule Injection

Trigger Examples

- *"Deliver stacks of exactly 3 parcels at a time to double the reward"*
 - *"Every time you deliver in $(x1,y1)$ you get 0 pts"*
-
- **RuleStore**: Defines a condition, multiplier (m), and offset (a)
 - **Effect**: Dynamically scales and shifts the BDI score for any desire matching the condition
 - **Special case**: The **Stack rule** automatically elevates **Explore** to priority 1

Coordination: Position Sharing

- **Beacons:** each agent broadcasts position at fixed interval
- Ensures up-to-date peer positions beyond sensing radius
- Needed for coordination when agents are outside each other's sensing radius

Coordination: 2-Phase Commit

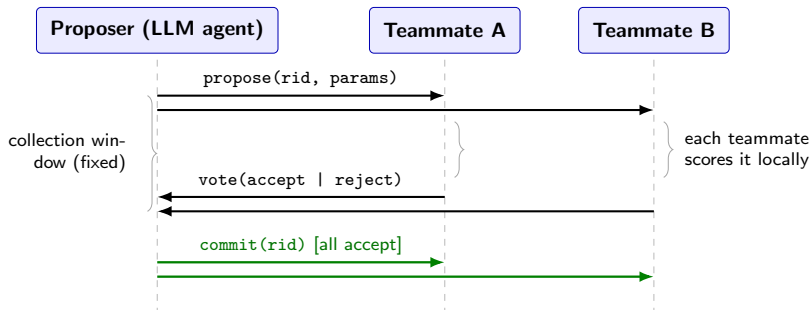
Trigger Example

- *“Move both agents to the neighborhood of position (x,y) within a maximum distance of 3, and have them wait for each other. You will receive 500pts.”*

Used for complex coordination tasks (commit together or not at all):

- **Phase 1 - Propose:** proposer broadcasts candidate hold tiles; each agent scores them
- **Vote:** *accept* if the hold outweighs current desires; *reject* if current desires are better
- **Phase 2 - Decide:** unanimous accept $\rightarrow$ commit (all inject); any miss or reject $\rightarrow$ abort

Coordination: 2-Phase Commit



Cooperation: Motivation

Origin: Admin Directive

"If a parcel is initially picked up by one agent and later delivered by the other agent, you will receive a 200 points bonus."

- A one-shot *check-and-coordinate* is insufficient: requires dynamic pickups, meeting tile computations, and joint commitments.
- ⇒ Solution: Dedicated **roles** with specific behaviours, agreed upon via a **2-phase commit** (2PC).
- Generalised into a reusable framework of **strategies** and **roles**.

Cooperation: Generalisation

- A **cooperation round** opens every Δt seconds (or immediately upon a mission directive).
- Agents request teammates' beliefs to build a **shared compact context**:

Context State

- **Roster**: Positions, payloads, sensed enemies.
 - **Map Geometry**: Spawn/delivery clusters, separation, corridor flags.
 - **Performance**: Average score/round per strategy (sliding window).
- ⇒ The **LLM** evaluates this context to select the optimal shared strategy for the team.

Cooperation: Strategies

- ZONAL_RELAY: One agent sweeps the spawn region and drops at a fixed midpoint; the companion ferries parcels to delivery.
- OPPORTUNISTIC: Agents forage independently and hand off parcels dynamically upon pickup.
- NONE: Fully autonomous baseline (no cooperation).

Cooperation: Roles

- Each strategy assigns every agent a specific **role**.
- Standard BDI desire generation is *blocked*.
- A role-specific **FSM** generates desires dynamically based on the shared context and internal state.
- Meeting points and positioning are computed *deterministically* from the map's geometry.

Cooperation: Handpass

Both strategies culminate in a **handpass**:

- ZONAL_RELAY:
The exchange loops indefinitely at the designated midpoint.
- OPPORTUNISTIC:
The agent acquiring a parcel proposes a handpass via 2PC.

[Collaboration demo]

Thank you for your attention!